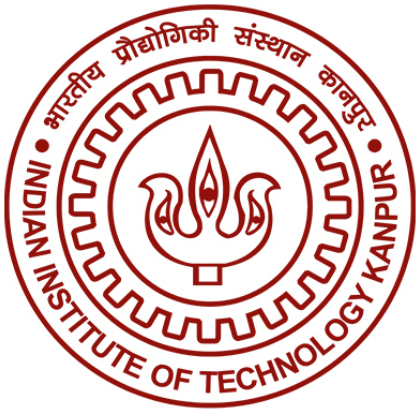


SPA617: RADIO ASTRONOMY LAB

Antenna Characterization

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Lab Report

Aim

To characterize the given antenna by measuring its beam pattern in both co-polarization and cross-polarization configurations.

Objectives

(a) To determine the antenna operating frequency and return loss from VSWR measurements.

(b) To measure the effect of distance on the power received by the antenna.

(c) To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

(d) To determine the half-power beamwidth and estimate the directivity of the antenna.

Theory

An antenna is a transducer that converts electrical power into electromagnetic waves and vice-versa. It can be used either as a transmitting antenna or as a receiving antenna. A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength ($\lambda/2$), with a feed line connected between two arms. It radiates most strongly in directions perpendicular to the wire and has very little radiation along the wire.

A monopole antenna is approximately half of a dipole mounted above a conducting ground plane. Its length is approximately one-quarter wavelength ($\lambda/4$). The ground acts like a mirror, making it behave like a full dipole. A far-field antenna region is the space far away from an antenna where EM waves act as flat plane waves. It is measured as $\max(3\lambda, 2d^2/\lambda)$, where λ = wavelength, d = dipole length. In the far-field region, the received power approximately follows the inverse-square law, $P \propto 1/r^2$ 'P' is received power and 'r' is separation between the transmitting and receiving antennas.

Polarization describes the orientation of electric field vector of an electromagnetic wave. For a linearly polarized wave, the electric field oscillates in a fixed direction perpendicular to the direction of propagation. Maximum signal is received when the receiving antenna is aligned with the polarization of the incident wave. We used co-polarized and cross-polarized configurations for our experiment.

In the co-polarized configuration, the transmitting and receiving antennas have aligned polarization, while in cross-polarized configuration, they are oriented perpendicular to each other as shown in the figure. The directional response of the horn antenna was measured in the E-plane and H-plane, defined according to the orientation of the sensing element. E-plane and H-plane are shown in the figure below. Rotating the antenna in these planes enabled the angular variation of the received signal to be measured and the corresponding antenna beam patterns to be characterized. Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. It is related to the beam solid angle 'Ω' as $D = (4\pi / \Omega)$. In practical, the beam solid angle can be approximated using the half-power beam widths in the E-plane (θ_E) and H-plane (θ_H), as, $D \approx 4\pi / (\theta_E \theta_H)$. Beam pattern of an antenna represents the variation of its radiated or received power with respect to direction. It typically consists of a main lobe, corresponding to the direction of maximum radiation, along with possible side lobes and a back lobe. Experimentally, the beam pattern can be determined by rotating the receiving antenna through different angles and recording the corresponding received power.

The half-power beam width (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value, corresponding to the −3 dB points. Small HPBW indicates a narrow and focused beam, whereas, a large HPBW indicates a broad beam.

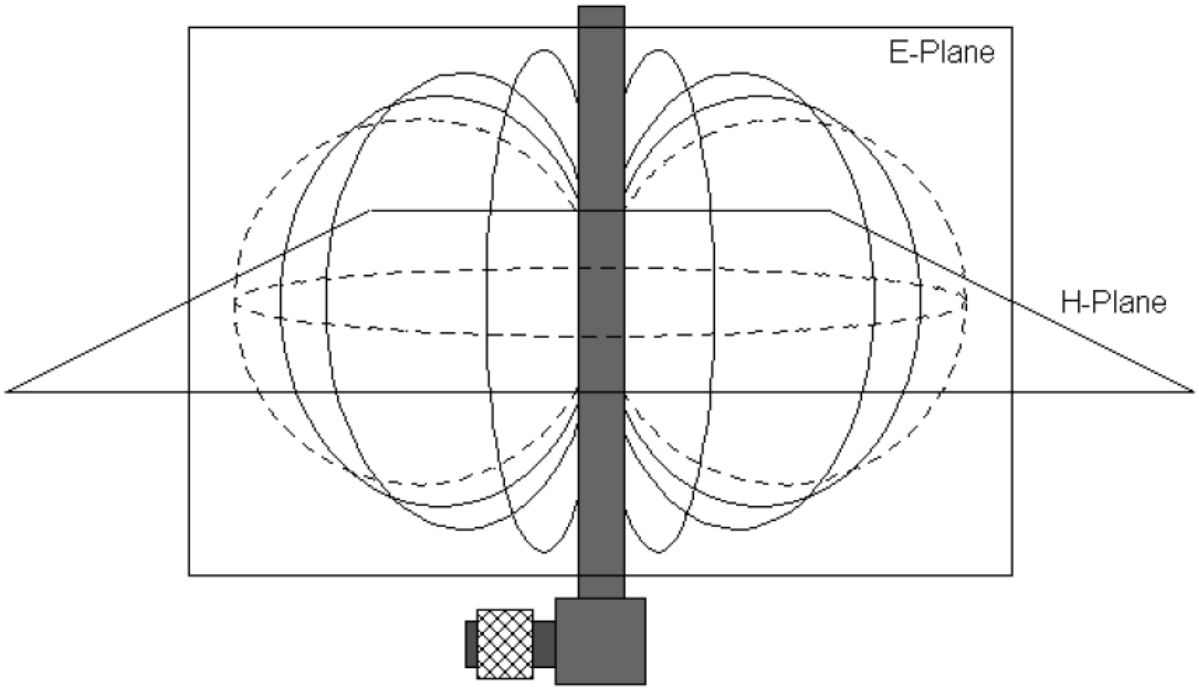


Figure: E-Plane and H-Plane

Co-Polarization and Cross-Polarization:

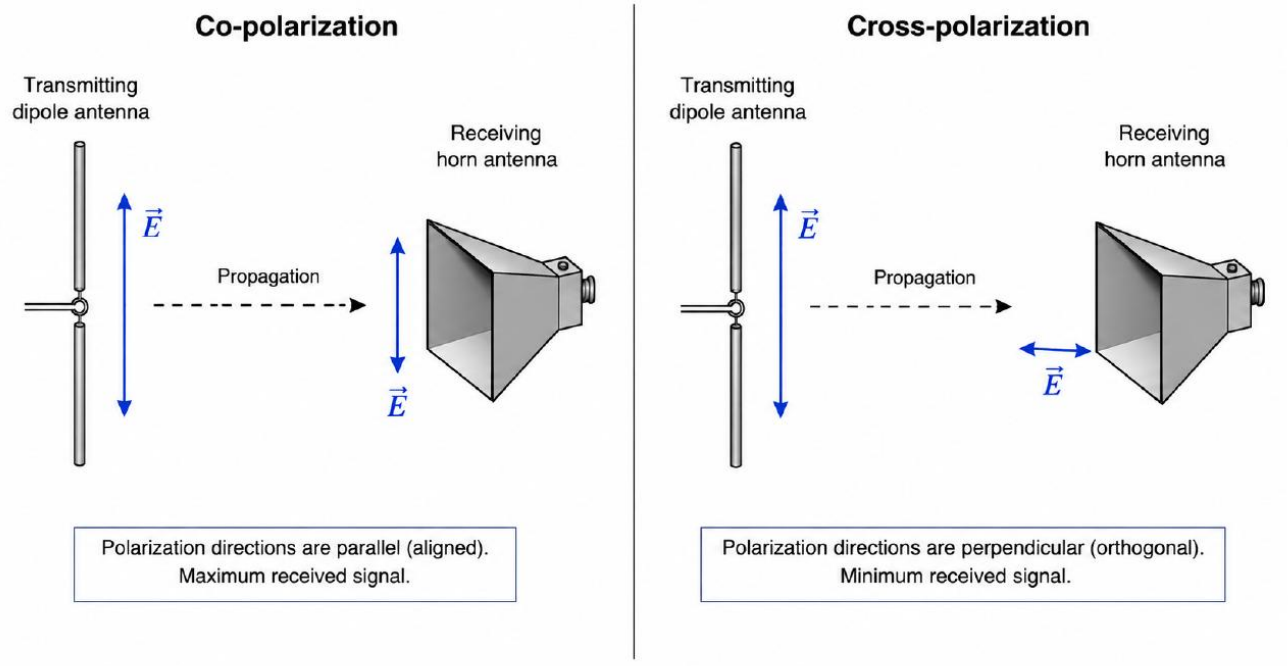


Figure: Co-polarisation (Left) and Cross-polarization (Right) configuration

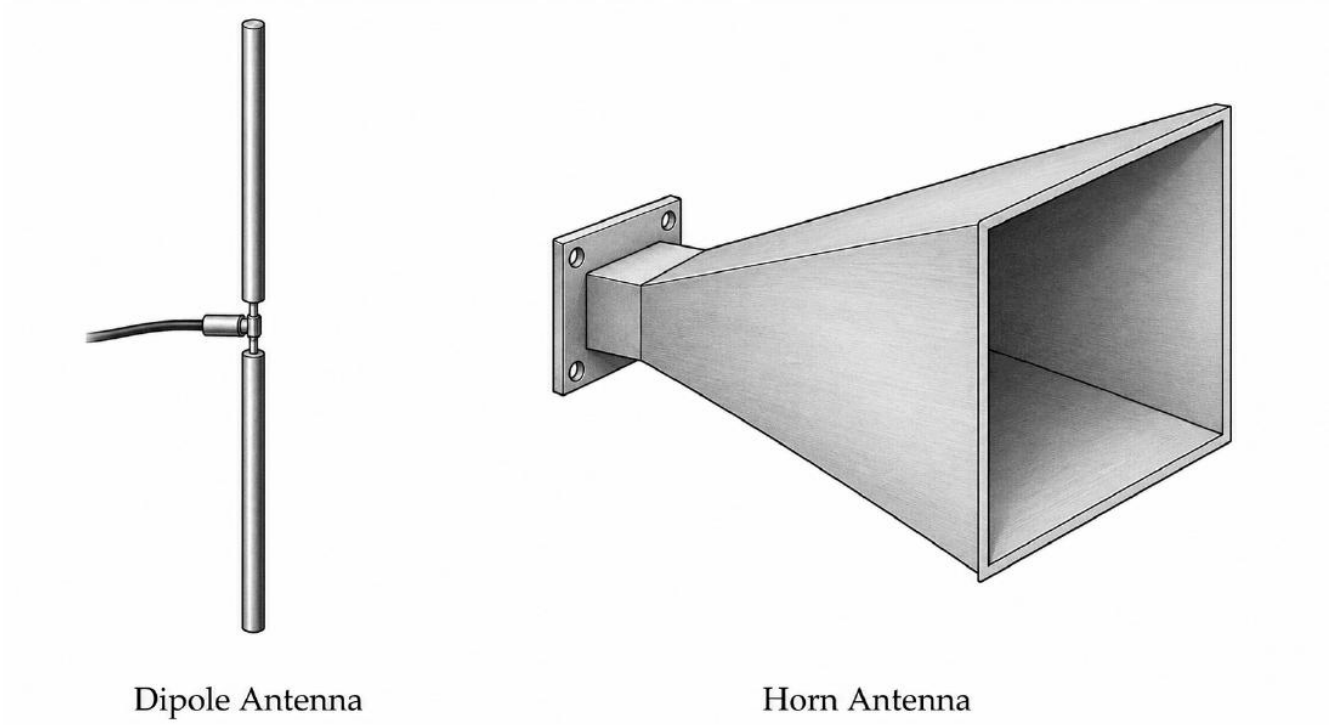


Figure: Dipole and Horn antennas

Components Required

(a) Horn antenna.

(b) Dipole antenna.

(c) RF Signal generator.

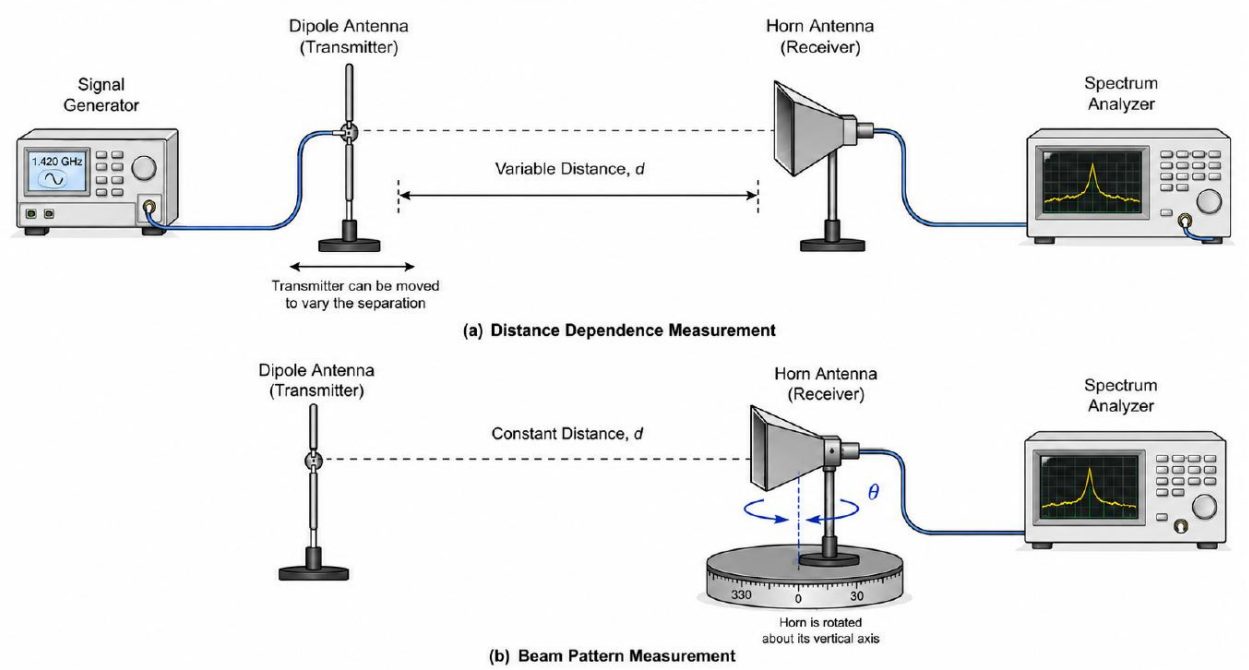
(d) Spectrum analyzer.

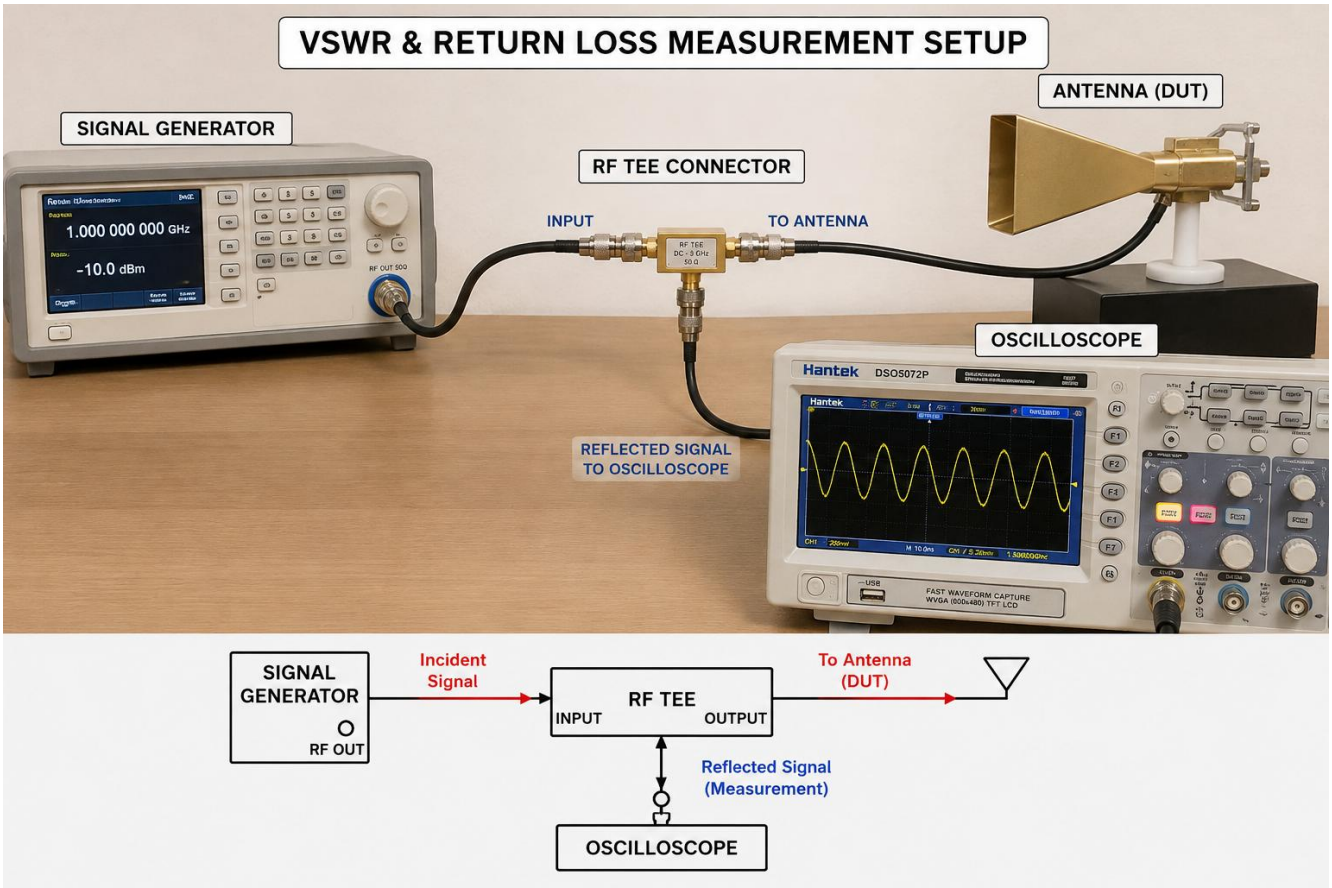
(e) Measuring tape

(f) Angular measurement scale.

Experimental Setup

Connect the transmitting antenna to the signal generator and the receiving antenna to the spectrum analyzer. Keep antennas at same height and align them appropriately. By varying the separation 'd' between the two antennas, we checked the dependence of received power with distance, while keeping their orientation same. For the beam-pattern measurements, the antenna separation was fixed and the horn antenna was rotated about its vertical axis by an angle θ. Angular position and received power were recorded at each distance with the help of spectrum analyzer.


Dipole Antenna
(Transmitter)
Horn Antenna
(Receiver)
Signal
Generator
Spectrum
Analyzer
1.420 GHz
Variable Distance, d
Transmitter can be moved
to vary the separation
(a) Distance Dependence Measurement
Dipole Antenna
(Transmitter)
Horn Antenna
(Receiver)
Spectrum
Analyzer
Constant Distance, d
θ
330
0
30
Horn is rotated
about its vertical axis
(b) Beam Pattern Measurement


VSWR & RETURN LOSS MEASUREMENT SETUP
SIGNAL GENERATOR
ANTENNA (DUT)
RF TEE CONNECTOR
INPUT
TO ANTENNA
1.000 000 000 GHz
-10.0 dBm
OSCILLOSCOPE
Hantek
DSOS072P
REFLECTED SIGNAL
TO OSCILLOSCOPE
SIGNAL
GENERATOR
RF OUT
Incident
Signal
RF TEE
INPUT
OUTPUT
To Antenna
(DUT)
Reflected Signal
(Measurement)
OSCILLOSCOPE

Observations and Calculations

Experiment was done by keeping the frequency at 1.42 GHz and output power level 20 dBm. The operating conditions for both the transmitting and receiving instruments were same. Then the separation distance and angular orientations were varied and noted the power changes. The recorded received-power values for the different experimental configurations are shown in the table below. For the distance dependent values, the separation between the transmitting dipole and receiving horn was varied while maintaining constant instrument settings.

Table 1: Power recieved vs angle (Co-polarization)

Angle	Power(dBm) for E plane	Power(dBm) for H plane
0	-36.40	-45.24
30	-33.27	-48.37
60	-37.80	-53.92
90	46.20	-62.62
120	-45.01	-55.09
150	-42.97	-51.50
180	-37.32	-52.32
210	-41.96	-43.93
240	-50.20	-48.85
270	-49.26	-57.14
300	-39.43	-52.72
330	-33.42	-45.95
360	-32.00	-45.22

Table 2: Power received vs angle (Cross-polarization)

Angle	Power(dBm) for E plane	Power(dBm) for H plane
0	-49.54	-41.47
30	-47.50	-43.32
60	-50.77	-40.40
90	-60.11	-43.09
120	-59.44	-42.33
150	-54.42	-47.71
180	-59.51	-41.73
210	-54.90	-47.40
240	-58.32	-59.33
270	-65.46	-49.34
300	-54.40	-44.29
330	-48.93	-40.96
360	-50.84	-40.97

Table 3: Distance vs power recieved table

Length	Power (dBm)	Received Power (dBm)	$20log_{10}(Length)$
217	20	-28.80	46.73%
343	20	-26.50	50.71%
418	20	-37.60	52.42%
495	20	-34.55	53.89&
572	20	-40.57	55.15%
694	20	-46.90	56.83%
726	20	-28.70	57.22%
803	20	-33.63	58.09%

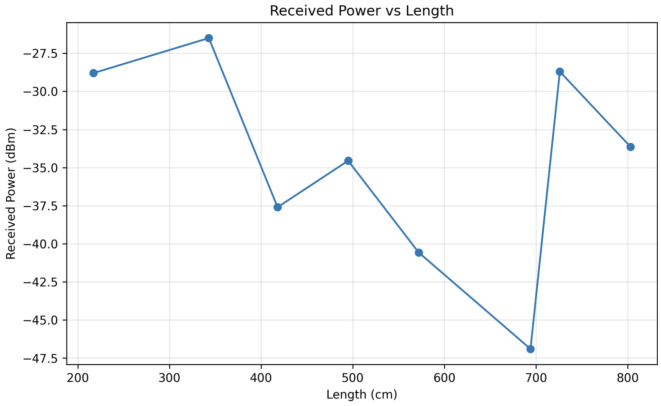
Table 4: Far-Field Table

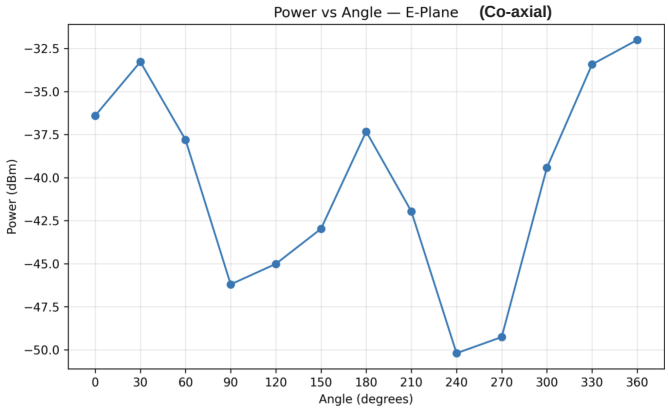
Antenna type	Maximum dimension of horn (cm)	Wavelength (cm)	Far-Field Distance (cm)
Horn	35	12.13	201.98 cm
Dipole	16	32	96 cm

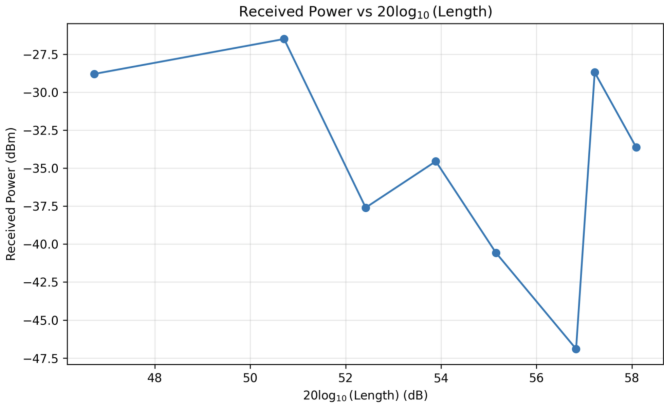
Table 5: Oscilloscope reading vs frequency

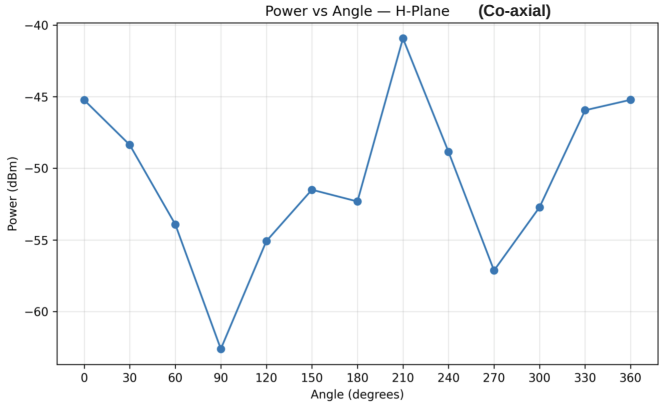
Ref. frequency or operating frequency (MHz)	Vmax of oscilloscope (volts)	Vmin of oscilloscope (mvolts)	Comments
103	1.75	700	
160.530	1.07	368.04	Corresponding frequencies are, 170.53MHz and 149 MHz

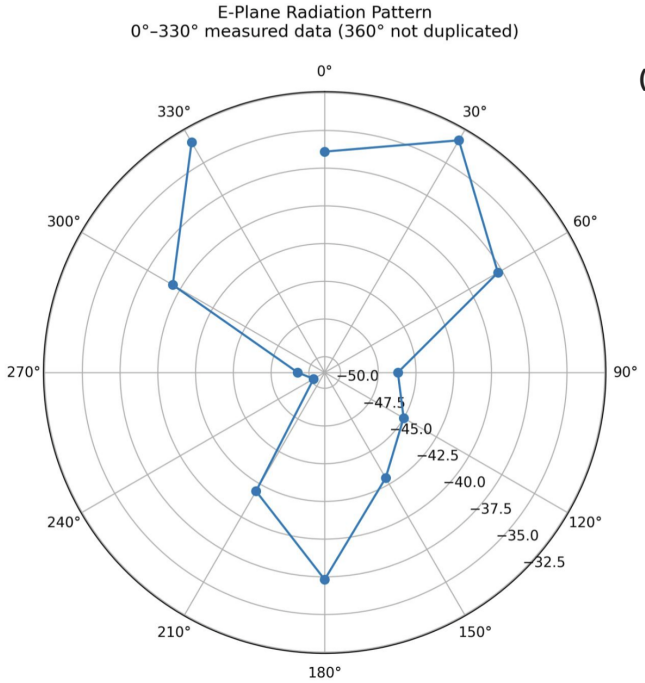
Graphs:



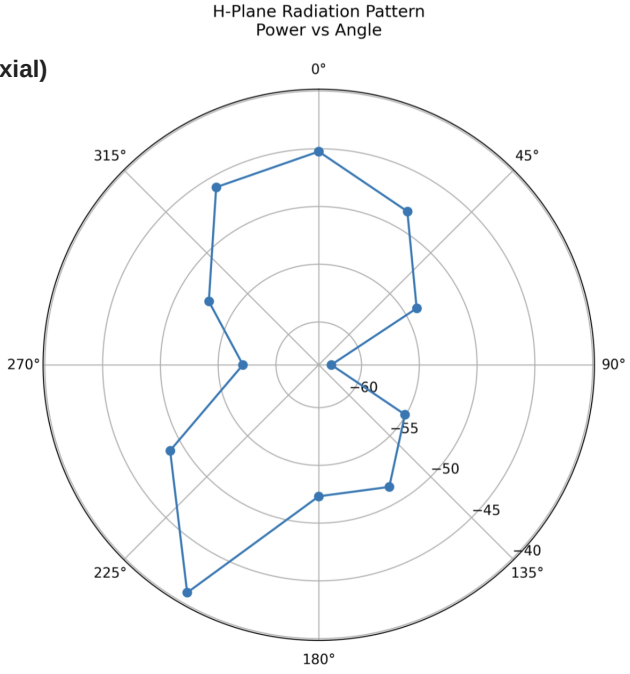


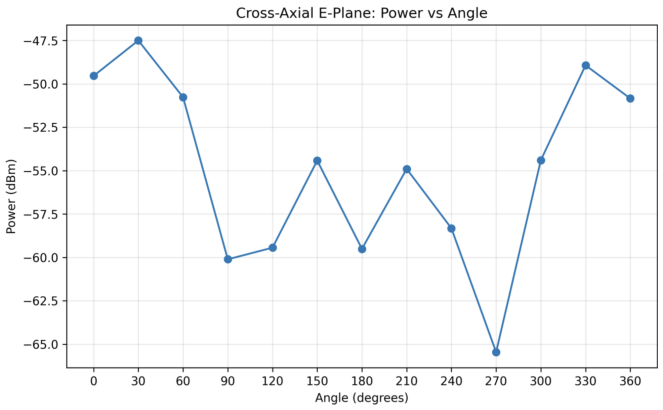


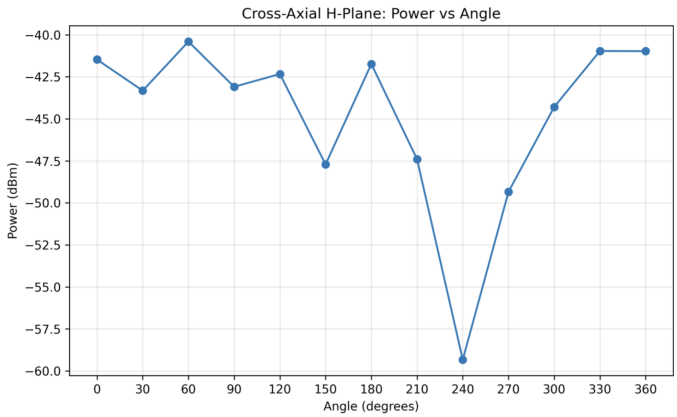


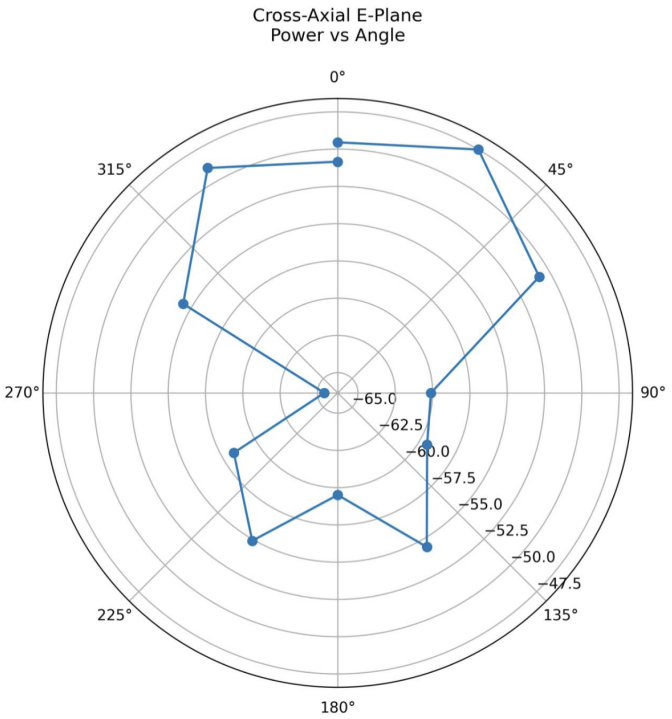


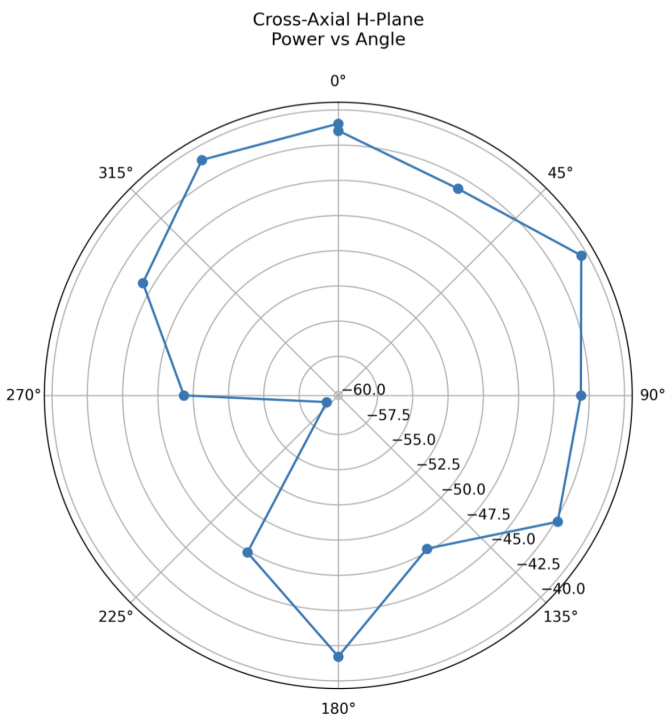
(Co-axial)











Measured length of pole of horn antenna = 4.5 cm = $\lambda/4$. So, λ = 18 cm.

Far-field distance = max(3λ, $2d^2/\lambda$). For Horn antenna, d = 35 cm, λ = 12.1 cm. So, far-field distance = 202 cm.

Maximum received power in the E-plane 'P(max, E)' = -32 dBm.
Half-power level corresponds to a decrease of approx. 3 dBm from the maximum received power.
So, P(-3 dBm, E) = P(max, E) - 3 dBm = -35 dBm, it's coming between 0^0 and 30^0, as −36.40 dBm and −33.27 dBm.
From linear interpolation, $\theta(E,L) = 0 + (30 - 0)x(-35 + 36.40)/(-33.27 + 36.40) \approx 13.42$ degrees.

Next -3dBm crosses between 300◦ and 330◦ as -39.43 dBm and -33.43 dBm.
$\theta(E,R) = 300 + (330 - 300)\,)x(\,-35 + 39.43)/(\,-33.43 + 39.43) \approx 322.15$ degrees.
So, $\theta(E,R) = 360 - 322.15 = 37.85$ degrees.
Therefore, E-plane half-power beam width (HPBW_E) = $\theta(E,R) - \theta(E,L) = 24.4$ degrees = 0.43 rad.

Maximum received power in H-plane 'P(max, H)' = -43.93 dBm.
Similarly, P(-3 dBm, H) = P(max, H) - 3 dBm = -46.93 dBm. It's coming between 0 degree and 30 degree, as −45.24 dBm and −48.37 dBm.
So, $\theta(H,L) = 0 + (30 - 0)x(\,-46.93 + 45.24)/(\,-48.37 + 45.24) \approx 16.20$ degrees.

Next -3 dBm crosses between 300◦ and 330◦ as -52.72 dBm and -45.95 dBm.
So, $\theta(H,R) = 300 + (330 - 300)\ x\ (\,-46.93 + 52.72)/(\,-45.95 + 52.72) \approx 360 - 325.66 = 34.34$ degrees.
Therefore, H-plane half-power beam width (HPBW_H) = $\theta(H,R) - \theta(H,L) = 18.14$ degrees = 0.32 rad.

Directivity = $(4\pi / (0.43 \times 0.32)) = 91.32$.
It shows that the antenna has preference for certain directions.

Quantity	**Sample value**
Operating frequency	1.42 GHz
Minimum VSWR	2.5
Return loss	12.67 db
E-plane HPBW	24.43°
H-plane HPBW	18.14°
Estimated directivity	91.32
Reflection Coefficient	0.43

Minimum VSWR = V(max) / V(min) = (1.75/0.7) = 2.5 . It gives frequency of best impedance matching.

Reflection Coefficient = (2.5-1)/(2.5+1) = 0.43 .
Return loss = $-20 \log |\Gamma|$ db = $-20 \log 0.43$ = -12.67 db.
Lower reflection coefficient represents higher return loss. Co-polarized pattern is stronger than the cross-polarized pattern. Received power decreases with antenna separation.

Conclusions

Some uncertainties:

- Reflection from surrounding.
- Antenna alignment.
- Measuring distance.
- Measuring angular-position.
- Instrumental error.

We didn't get inverse sq. law power variation with distance. Maybe it's due to reflection from nearby walls. Reflection coefficient is not zero due to calculated operating frequency is not same with real operating frequency. Received power decreases with distance, and co-polarization produces a stronger response than cross-polarization.